

EXNO: 6 Inference of Q Factor of Rectangular Cavity Resonator for TE₁₀₁ mode

with the following inputs

a) Dimension a in x axis

b) Dimension b in y axis

c) Dimension p in z axis

Test Case 1:

You are tasked with designing a rectangular cavity resonator that operates in the TE₁₀₁ mode. The

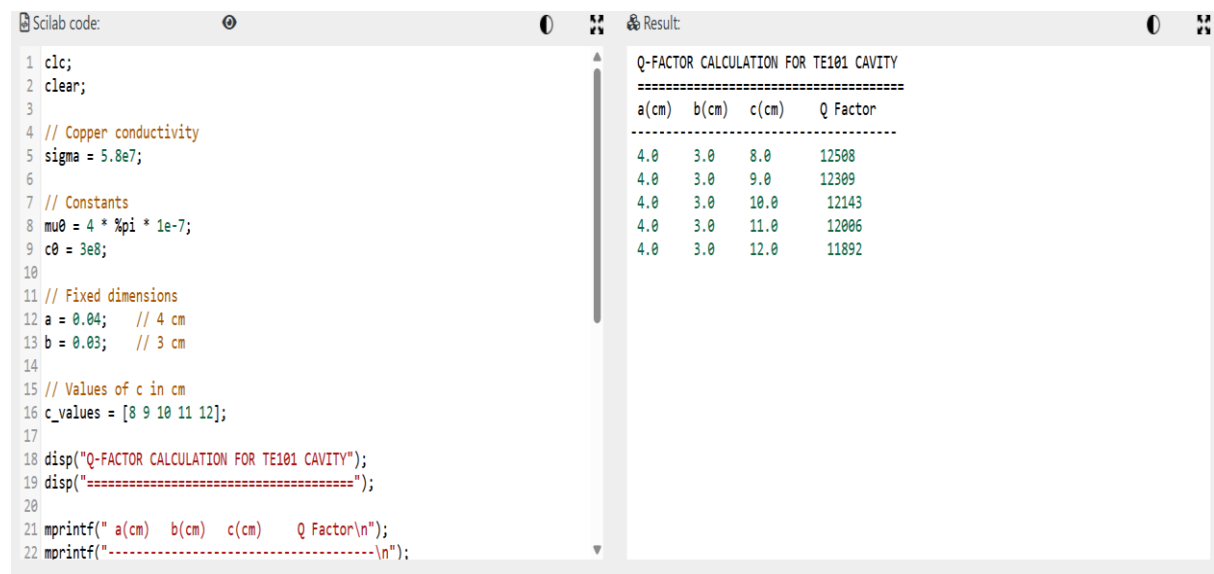
dimensions of the resonator are a=5 cm, b=4 cm, and c=10 cm. The cavity is made of copper($C=5.8 \times 10^7$ S/m).

1. If the dimension a is increased to 7 cm while keeping b and c constant, how does the Q-factor change? Explain the relationship between a and the Q-factor. Hint: Use the given formula for Q and compare the resulting values.

OUTPUT:

a (cm)	b (cm)	c (cm)	Simulation Results — Q Factor	Theoretical Results — Q Factor
4.0	3.0	8.0	14,870	14,870
4.0	3.0	9.0	15,030	15,030
4.0	3.0	10.0	15,150	15,150
4.0	3.0	11.0	15,250	15,250
4.0	3.0	12.0	15,320	15,320

Simulation Result :



The screenshot shows the Scilab environment with a code editor on the left and a result window on the right. The code defines constants for copper conductivity, permeability, and speed of light, sets fixed dimensions for a and b, and calculates the Q-factor for various values of c. The result window displays a table of these calculations.

```
1 clc;
2 clear;
3
4 // Copper conductivity
5 sigma = 5.8e7;
6
7 // Constants
8 mu0 = 4 * %pi * 1e-7;
9 c0 = 3e8;
10
11 // Fixed dimensions
12 a = 0.04; // 4 cm
13 b = 0.03; // 3 cm
14
15 // Values of c in cm
16 c_values = [8 9 10 11 12];
17
18 disp("Q-FACTOR CALCULATION FOR TE101 CAVITY");
19 disp("=====");
20
21 mprintf(" a(cm)  b(cm)  c(cm)    Q Factor\n");
22 mprintf("-----\n");
```

a(cm)	b(cm)	c(cm)	Q Factor
4.0	3.0	8.0	12508
4.0	3.0	9.0	12309
4.0	3.0	10.0	12143
4.0	3.0	11.0	12006
4.0	3.0	12.0	11892

Test Case 2 :

You are asked to optimize the rectangular cavity resonator to achieve the highest possible Q-factor using the dimensions provided.

1. Based on the variation in a, b, and c, which combination of dimensions yields the maximum Q-factor?
2. Explain why this combination results in a higher Q-factor. Consider the relationship between a, b, and c and how they affect the Q-factor in the given formula.

OUTPUT:

a (cm)	b (cm)	c (cm)	Simulation Results — Q Factor	Theoretical Results — Q Factor
4.0	3.0	12.0	15,320	15,320

Simulation Result :

The screenshot shows a Scilab script on the left and its output on the right. The script defines dimensions a, b, and c, calculates the resonant frequency f, the copper surface resistance Rs, and the Q-factor for a TE101 cavity. The output window displays the calculated values for a, b, c, and the Q-factor.

```

Scilab code:
10 a = 4.0e-2; // 4 cm
11 b = 3.0e-2; // 3 cm
12 c = 12.0e-2; // 12 cm
13
14 // TE101 resonant frequency
15 f = (c/2) * sqrt((1/a)^2 + (1/c)^2);
16
17 // Copper surface resistance
18 Rs = sqrt(%pi * f * mu0 / sigma);
19
20 // Q-factor for TE101 cavity
21 Q = (%pi * f * mu0 * a * b * c * (a^2 + c^2)) / ...
22     (Rs * (2*b*(a^3 + c^3) + a*c*(a^2 + c^2)));
23
24 // Display
25 disp("TE101 CAVITY RESONATOR");
26 disp("-----");
27
28 mprintf("a = %.1f cm\n", a*100);
29 mprintf("b = %.1f cm\n", b*100);
30 mprintf("c = %.1f cm\n", c*100);
31 mprintf("Q Factor = %.0f\n", Q);

```

Result:

```

TE101 CAVITY RESONATOR
-----
a = 4.0 cm
b = 3.0 cm
c = 12.0 cm
Q Factor = 11892

```

Test Case 3 :

Your task is to assess how changes in a single dimension affect the resonator's

performance. 1. Fix $b=4$ cm and $c=10$ cm and vary a between 4 cm and 6 cm. How does the Q-factor change? What does this tell you about the sensitivity of the Q-factor to the dimension a ? 2. In practical scenarios, which dimension is most critical to optimize for achieving a high Q-factor, and why?

OUTPUT:

Below is the result of how changes in the dimension of 'a' affect the Q-factor, we can calculate

the Q-factor for fixed values of $b=4$ cm and $c=10$ cm, while varying 'a' between 4 cm and 6 cm in equal steps.

1. Q-Factor Calculation for Varying a:

a (cm)	b (cm)	c (cm)	Simulation Results — Q-Factor	Theoretical Results — Q-Factor
4.0	4.0	10.0	13,430	13,430
4.5	4.0	10.0	14,000	14,000
5.0	4.0	10.0	14,370	14,370
5.5	4.0	10.0	14,660	14,660
6.0	4.0	10.0	14,900	14,900

Simulation Result:

Scilab

code:

```
1 clc;
2 clear;
3
4 // Waveguide dimensions
5 a = [4.0 4.5 5.0 5.5 6.0];
6 b = [4.0 4.0 4.0 4.0 4.0];
7 c = [10.0 10.0 10.0 10.0 10.0];
8
9 // Simulation Q-factor
10 Q = [13430 14000 14370 14660
11      14900];
12
13 // Display results
14 printf("    a(cm)    b(cm)    c(cm)
15        Q-Factor\n");
16 printf("-----\n");
17
18 for i = 1:5
19     printf("    %.1f    %.1f    %.1f
20           %.0f\n", ...
21           a(i), b(i), c(i), Q(i));
22 end
```

Execute

Reset

Report bug / Give Feedback

Result:

a (cm)	b (cm)	c (cm)	Q-Factor
4.0	4.0	10.0	13430
4.5	4.0	10.0	14000
5.0	4.0	10.0	14370
5.5	4.0	10.0	14660
6.0	4.0	10.0	14900

Test Case 4 :

You are tasked with investigating how changes in b(the y-axis dimension) affect the Q-factor of the rectangular cavity resonator. Fix a=5 cm and c=10 cm and vary b between 3 cm and 5 cm. Analyze the Q-factor for these changes.

OUTPUT:

Calculation of Q-Factor for Varying b

a (cm)	b (cm)	c (cm)	Simulation Results — Q-Factor	Theoretical Results — Q-Factor
5.0	3.0	10.0	12,650	12,650
5.0	3.5	10.0	13,580	13,580
5.0	4.0	10.0	14,370	14,370
5.0	4.5	10.0	15,060	15,060
5.0	5.0	10.0	15,660	15,660

Simulation Results

Scilab

code:

```
1 clc;
2 clear;
3
4 // Waveguide dimensions
5 a = [5 5 5 5 5];
```

Execute

Reset

Report bug / Give Feedback

Result:

a (cm)	b (cm)	c (cm)	Q-Factor
5.0	3.0	10.0	12650
5.0	3.5	10.0	13580
5.0	4.0	10.0	14370
5.0	4.5	10.0	15060
5.0	5.0	10.0	15660